

Pokhara University Affiliate

LA GRANDEE INTERNATIONAL COLLEGE

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A Project Proposal

on

‘Rice Management System’

Submitted to

Bachelor of Computer Application (BCA) Department

LA GRANDEE International College

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Application under Pokhara University

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1. Introduction

Rice is one of the most important staple crops in Nepal. Managing rice production, procurement, storage, and distribution efficiently is critical for ensuring food security and economic stability. Traditional methods of rice inventory and stock management, which rely heavily on manual record keeping, are to human errors, data redundancy, and inefficiency.

The Rice Management System is a software-based solution designed to automate and digitize the management of rice-related operations including stock tracking, supplier management, sales recording, and report generation. The system aims to provide an efficient, reliable, and user-friendly platform for rice mills, warehouses, cooperatives, and distribution centers to manage their operations seamlessly.

This project is developed as part of the Bachelor of Computer Application (BCA) program under Pokhara University. The system will be a desktop-based application providing a centralized database to manage all aspects of rice management from procurement to final distribution.

1.1 Aim And Scope

The primary aim of this project is to design and develop a Rice Management System that streamlines the operations of rice storage, quality checking, and distribution for businesses and cooperatives dealing with rice commodities.

The scope of the system includes:

- Managing supplier and buyer records
- Tracking rice stock levels including varieties, quantities, and grades
- Recording purchase and sales transactions
- Generating inventory and financial reports
- Monitoring warehouse stock in real time

1.2 Objective

The following are the key objectives of the Rice Management System:

1. To develop a centralized digital platform for recording and managing rice stock and transactions.
2. To automate stock tracking and generate real-time inventory reports.
3. To maintain records of suppliers, buyers, and employees efficiently.
4. To reduce manual errors and improve data accuracy in rice operations.
5. To generate profit, loss, and transaction reports for business decision making.

2. Study Of Previous System

The development of this system, rice management operations in most small to medium-sized businesses in Nepal were carried out using manual record-keeping systems such as paper ledgers, log books, and basic spreadsheet tools like Microsoft Excel. While these methods have served businesses over the years.

A review of the existing manual and semi-digital practices revealed the following characteristics of the previous system:

2.1 Manual Record Keeping

Rice stock entries, purchase records, sales figures, and supplier details were maintained in physical registers. This made data retrieval time-consuming and loss due to damage or misplacement of records.

2.2 Use Of Basic Spreadsheets

Some businesses used Microsoft Excel for stock tracking. While better than paper records, spreadsheets lacked automated calculations, real-time stock updates, multi-user access control, and report generation capabilities.

2.3 Absence Of Centralized Database

Information about suppliers, buyers, employees, and transactions were stored in different files or registers without any centralized system. This resulted in data inconsistency and duplication.

2.4 Lack Of Real-Time Monitoring

There was no mechanism for monitoring stock levels in real time. Low stock warnings were based purely on periodic manual counts, leading to delays in procurement and sometimes stock shortages.

3. Current Problem And Proposed Solution

3.1 Current Problems

The following major problems have been identified in the current system of rice management in small and medium enterprises:

- Manual entry of stock data leads to frequent calculation errors and data inaccuracies.
- No real-time visibility of stock levels, causing procurement delays and overstocking.
- Lack of supplier and buyer management makes tracking business relationships difficult.
- Generating financial or inventory reports manually is time-consuming and unreliable.
- Absence of access control means unauthorized personnel can alter or delete important records.
- Data is scattered across multiple files and registers, making it difficult to audit or review.
- Risk of data loss due to damage, fire, or theft of physical records.

3.2 Proposed Solution

The proposed Rice Management System addresses all of the above problems through the following solutions:

- Entering stock data manually can cause mistakes and wrong calculations.
- There is no real-time stock update, which can lead to delays or extra stock.
- Managing supplier and buyer details is difficult.
- Making inventory and financial reports manually takes a lot of time.
- Anyone can change or delete records because there is no proper access control.
- Data is stored in different files and registers, so checking records becomes difficult.
- Physical records can be lost due to fire, theft, or damage.

4. High Level Requirements

4.1 Functional Requirements

The functional requirements define the core features and behaviors the system must support:

4.1.1 User Authentication

- Users must be able to log in with a unique username and password.

4.1.2 Stock Management

- The system shall allow add, edit, and delete rice stock records.
- Each stock entry shall include rice variety, quantity, grade, and price per unit.
- Real-time stock level updates after each purchase or sale.

4.1.3 Sales And Transaction Management

- The system shall record all sales transactions including buyer details, quantity, and price.
- Invoice generation for each completed transaction.

4.1.4 Report Generation

- The system shall generate inventory reports, sales reports, purchase reports, and profit/loss reports.

4.1.5 Low Stock Alert

- The system shall automatically generate an alert when stock falls below a user-defined minimum level.

4.2 Non-Functional Requirements

The non-functional requirements define the quality attributes and constraints of the system:

4.2.1 Performance

- The system shall respond to user queries within 2 to 5 seconds under normal operating conditions.

4.2.2 Usability

- The interface shall be intuitive and require minimal training for users.

4.2.3 Reliability

- The system shall achieve at least 90% uptime during business hours.

4.2.4 Maintainability

- The codebase shall follow standard coding conventions for easy future maintenance.
- The system shall be modular to allow easy addition of new features.

4.3 Requirement Prioritization Table

Table 4.3.1 - Requirement Prioritization Table

S.N.	Requirement	Priority	Type	Category
1	User Authentication	High	Functional	Must Have
2	Stock Management Module	High	Functional	Must Have
3	Supplier Management	High	Functional	Must Have
4	Sales and Transaction Management	High	Functional	Must Have
5	Report Generation	Medium	Functional	Should Have
6	Low Stock Alert System	Medium	Functional	Should Have
7	Invoice Generation	Medium	Functional	Should Have
8	System Performance	Medium	Non-Functional	Should Have
9	User-Friendly Interface	Medium	Non-Functional	Should Have

5. Design

5.1 Er Diagram

The Entity-Relationship (ER) diagram illustrates the logical structure of the database and the relationships between key entities in the Rice Management System. The primary entities include User, Stock, Supplier, Buyer, Transaction, and Alert.

Key entities and their relationships are described as follows:

- User entity stores login credentials and role information for all system users.
- Stock entity records details of rice varieties, quantities, grades, and warehouse location.
- Supplier entity stores supplier information including name, contact, and address.
- Buyer entity maintains buyer records for sales transactions.
- Transaction entity links suppliers and buyers to stock entries through purchase and sales records.
- Alert entity stores low stock notifications with thresholds linked to stock items.

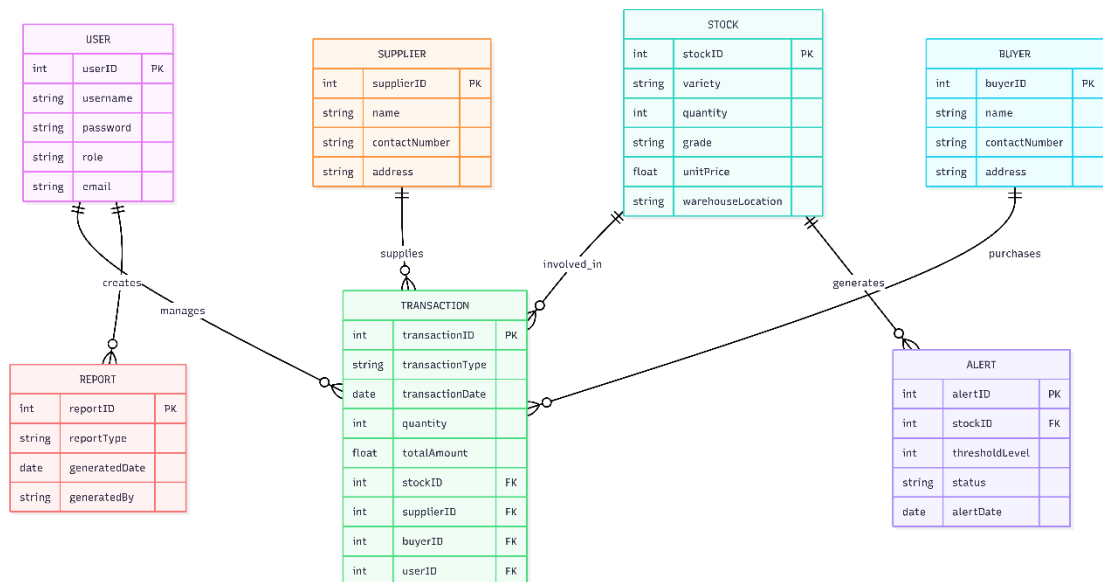


Figure 5.1.1 - Entity Relationship Diagram

5.2 Use Case

The Use Case diagram represents the interaction between actors (Admin, Manager, Staff) and the various functionalities of the Rice Management System.

Primary actors and their interactions include:

- Admin: Can perform all operations including user management, stock management, supplier management, sales management, and report generation.
- Manager: Can manage stock, transactions, suppliers, buyers, and view reports but cannot manage users.
- Staff: Can view stock, record sales transactions, and raise stock alerts.

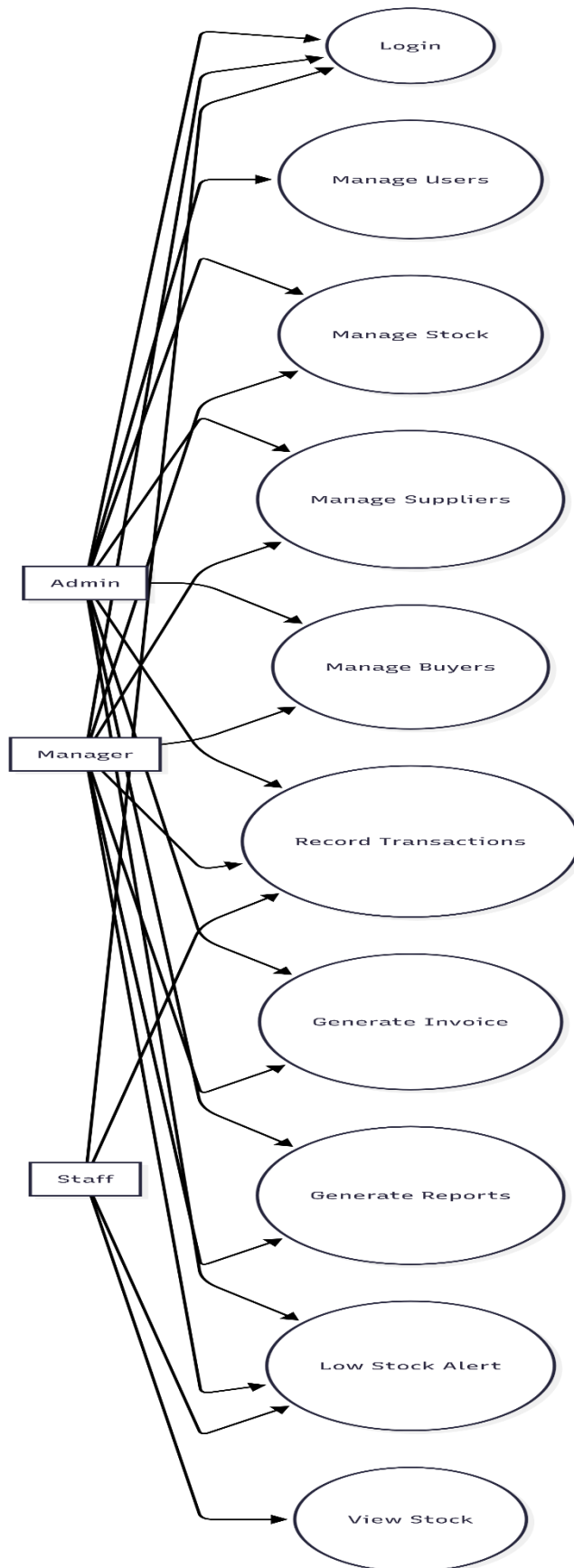


Figure 5.2.1 - Use Case Diagram

5.3 Dfd

The Data Flow Diagram (DFD) illustrates the flow of data within the Rice Management System at different levels of abstraction.

5.3.1 Level 0 Dfd (Context Diagram)

The Level 0 DFD shows the system as a single process with its external entities: Admin, Manager, Staff, Supplier, and Buyer. Data flows into the system from all actors and the system outputs reports, invoices, and alerts to the relevant actors.

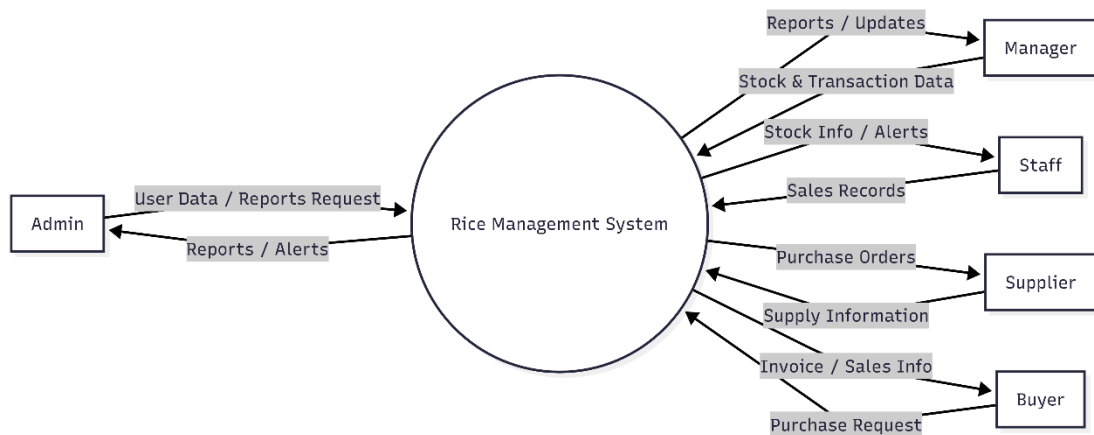


Figure 5.3.1 - Level 0 DFD (Context Diagram)

5.3.2 Level 1 Dfd

The Level 1 DFD decomposes the system into major sub-processes: Authentication, Stock Management, Supplier Management, Transaction Management, and Report Generation. Data stores such as the User Database, Stock Database, Transaction Database, and Supplier Database are also illustrated.

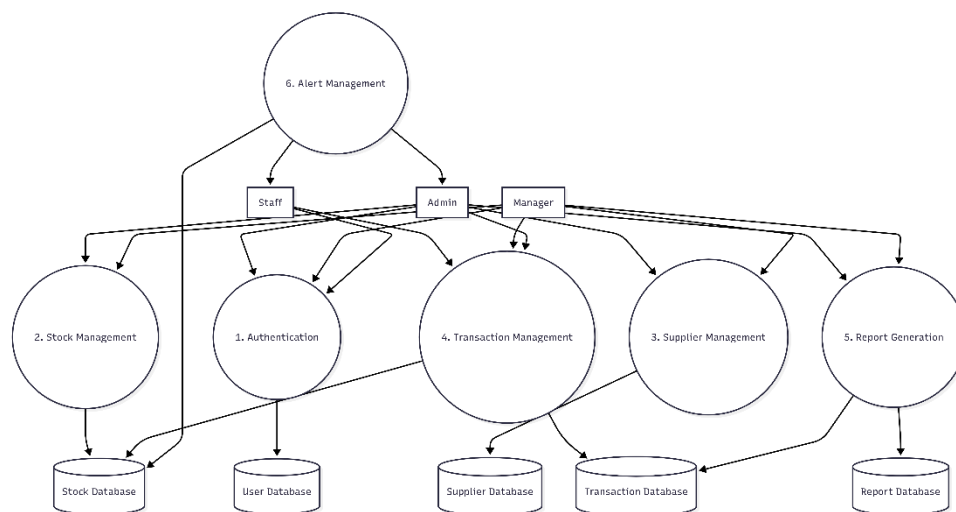


Figure 5.3.2 - Level 1 DFD

5.4 Uml Class Diagram

The UML Class Diagram represents the object-oriented structure of the Rice Management System, defining classes, their attributes, methods, and associations.

The primary classes in the system are:

- User: Attributes include userID, username, password, role, and email. Methods include login(), logout(), and updateProfile().
- Stock: Attributes include stockID, variety, quantity, grade, unitPrice, and warehouseLocation. Methods include addStock(), updateStock(), and deleteStock().
- Supplier: Attributes include supplierID, name, contactNumber, and address. Methods include addSupplier(), updateSupplier(), and removeSupplier().
- Transaction: Attributes include transactionID, type (purchase/sale), date, quantity, and totalAmount. Methods include recordTransaction() and generateInvoice().
- Alert: Attributes include alertID, stockID, threshold, and status. Methods include triggerAlert() and resolveAlert().
- Report: Attributes include reportID, reportType, generatedDate, and data. Methods include generateReport() and exportPDF().

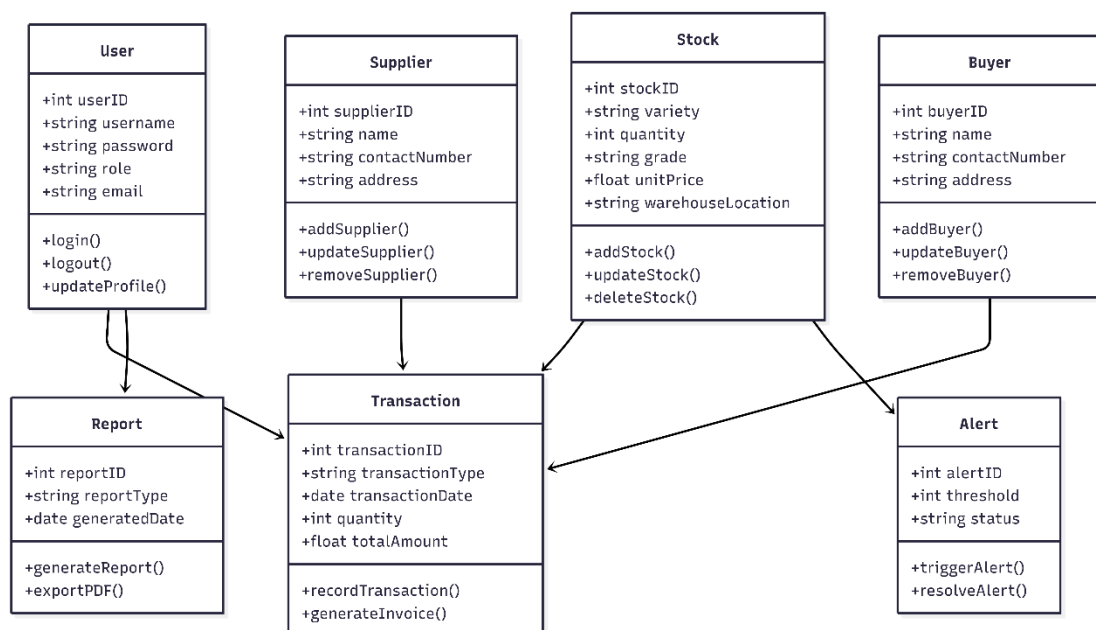


Figure 5.4.1 - UML Class Diagram

6. Detail Budget

6.1 Budget Breakdown

The following table presents the detailed budget estimation for the development of the Rice Management System. The budget covers software tools, hardware requirements, documentation, and miscellaneous expenses.

Table 6.1.1 - Budget Breakdown

S.N.	Items	Descriptions	Cost
1	Laptop / Computer Hardware	Any those run properly	30,000
2	Software Tools	Visual Studio Code	Free
3	JavaFX lib	OpenJFX – open source	Free
	Total Estimated Budget		30,000

7. Detail Timeline

The development of the Rice Management System is planned over a 45-day period organized into 7 weeks. The timeline is structured to ensure systematic progress from requirements analysis through to final testing and submission.

Table 7.1.1 - Project Timeline

Week	Dates	Phase / Activity	Deliverables / Milestones
Week 1	May 6-10, 2026	Requirement Analysis and Planning	Finalized requirements list, project plan, and proposal document
Week 2	May 11-17, 2026	System Design	ER Diagram, DFD, Use Case Diagram, UML Class Diagram completed
Week 3	May 18-24, 2026	Database Design	Database mainly used on .txt
Week 4	May 25-31, 2026	Core Module Development	User login, stock management, and supplier management modules developed
Week 5	Jun 1-7, 2026	Transaction and Report Module Development	Sales/purchase transaction module, invoice generation.
Week 6	Jun 8-14, 2026	GUI Development and Integration	Complete UI design, integration of JavaFX, alert module
Week 7	Jun 15-20, 2026	Testing, Bug Fixing, and Final Submission	System testing, bug fixing, documentation finalization, and project submission

8. Conclusion

The Rice Management System is a timely and necessary software solution aimed at addressing the inefficiencies and errors associated with manual rice management operations. Through the automation of stock tracking, supplier management, transaction recording, and report generation, the system will significantly improve the operational efficiency of rice mills, cooperatives, and distribution centers.

The system is designed using a structured software development approach, incorporating requirement analysis, system design, database design, module development, and testing phases.

This project will not only fulfill the academic requirement for the Bachelor of Computer Application (BCA) program under Pokhara University but will also produce a practical, deployable software solution with real-world applicability for agricultural businesses in Nepal. The development team is committed to delivering a well-tested, user-friendly, and maintainable system within the 45-day timeline.

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